Crime Hotspot Prediction System

Problem Statement Document

Final Year Minor Project

Department	B.Tech in Data Science
College	Swami Vivekananda University
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1. Introduction and Background

Crime affects how safe people feel while living, working and travelling in a city. Police departments record every reported incident with its time, type and location. Together these records form a large dataset that can show where and when crime is concentrated. Patrolling decisions are often made on experience alone. Data-driven methods can support them with evidence.

This project uses machine learning on past, geo-tagged crime records to find crime hotspots and to predict which areas are likely to have high crime next week. The result is shown through an easy web application.

2. Problem Statement

To design and build a data-driven system that identifies and predicts crime hotspots in a city from historical, geo-tagged crime records using machine learning, and presents the next-week prediction for every 500 m area through an interactive web application.

3. Objectives

- Analyze crime patterns by time (hour, day, month), place (district, grid cell, location type) and crime type.
- Identify crime hotspots using a heat map and spatial clustering (DBSCAN), and visualize them on interactive maps.
- Predict next-week crime counts and hotspot status for each area using regression and classification models (Random Forest, XGBoost).
- Evaluate the models with a time-based train/test split (80:20, 70:30 and 85:15) and compare them with simple baselines.
- Deploy the trained model as a Streamlit web application on the cloud so that anyone can view next-week hotspots.

4. Scope and Assumptions

- The project uses reported crime data only. Unreported crime is not captured.
- Crime is modelled at area level (about 500 m grid cells) and weekly level, not as predictions about individual people.
- The pipeline is built and validated on one city's open data. The same pipeline can be applied to another city if similar data becomes available.
- Hotspot risk is a decision-support score. It does not guarantee the safety of any place.
- A user login system is out of scope, because it does not contribute to the data-science objectives.

5. Dataset

To the best of our knowledge, publicly available Indian crime data is mostly aggregated at state or district level and does not include incident coordinates, so it cannot support hotspot mapping. We therefore chose a well-documented open dataset with incident-level coordinates to build and test the pipeline.

Item	Details
Source	City of Chicago Data Portal, Crimes - 2001 to Present (open data), downloaded as CSV
Period used	1 January 2022 to 31 December 2022
Size	240,512 records x 22 columns (above the 1 lakh row requirement)
Key columns	date, primary_type, description, latitude, longitude, district, beat, ward, community_area, location_description, arrest, domestic

6. Overall System Workflow

1. Data	2. Cleaning	3. Features	4. Hotspots	5. Prediction	6. Web app
Chicago crime records (CSV)	Missing values, duplicates, date format	Time features, grid cells, lag counts	Heat map and DBSCAN clusters	Regression and classification models	Streamlit app deployed on the cloud

7. Complete Project Roadmap (Start to End)

The table lists every stage of the project in order, the techniques used, and the current status. **All stages are completed.**

#	Stage	What is done (techniques)	Status
1	Problem definition and dataset selection	Defined the problem and objectives; surveyed open crime datasets; selected Chicago 2022 crime records because they include incident-level coordinates.	**Done**
2	Data collection	Downloaded the 2022 records as a CSV file from the City of Chicago data portal.	**Done**
3	Data loading and understanding	Loaded with pandas; checked shape, column types and first records.	**Done**
4	Data quality check	Counted missing values per column and checked duplicate rows.	**Done**
5	Data cleaning	Kept 12 useful columns; converted date to datetime; removed rows without coordinates and wrong-location rows; filled missing location_description with UNKNOWN; removed exact duplicates.	**Done**
6	Feature engineering	Created hour, day of week, month, weekend and night features from the date; created 500 m grid cell IDs from latitude and longitude.	**Done**
7	Exploratory data analysis (EDA)	Top crime types; crimes by hour, day and month; day-vs-hour heat map; location scatter plot; share of crimes held by the busiest grid cells; written observations for each chart.	**Done**
8	Hotspot detection	Interactive Folium heat map; DBSCAN clustering on latitude/longitude of the busiest cells; hotspot cluster map.	**Done**
9	ML dataset preparation	Built a weekly crime count per grid cell; added lag features (previous 1-4 weeks), 4-week rolling average, month and cell location; split by time (earlier weeks train, later weeks test).	**Done**
10	Baseline and regression models	Baselines (last week, 4-week average); Random Forest and XGBoost regressors to predict next-week crime count; metrics MAE, RMSE, R-squared.	**Done**
11	Hotspot classification	Labelled a cell as hotspot using a threshold taken from the training data only; Random Forest classifier with class weights; metrics F1 and ROC-AUC; feature importance.	**Done**
12	Model improvement	Compared three train/test splits; hyper-parameter tuning; compared grid sizes (250 m, 500 m, 1 km).	**Done**
13	Integration and saving files	Saved model settings, training data and next-week input files for the web application.	**Done**
14	Streamlit web application	Built the app with three tabs: City Hotspot Map, Check a Location and What-if Calculator.	**Done**
15	Cloud deployment	Deployed the Streamlit application on the cloud.	**Done**

#	Stage	What is done (techniques)	Status
16	Documentation and presentation	Final project documentation, presentation and viva preparation.	**Done**

Tools: Python, pandas, numpy, scikit-learn, XGBoost, matplotlib, seaborn, Folium, Plotly, Streamlit, Jupyter Notebook.

8. System Architecture of the Web Application

Layer	What it does	Tools
Presentation layer	Three tabs: City Hotspot Map (top-K risky areas, CSV download), Check a Location (nearest area, predicted crimes, hotspot probability, risk level) and What-if Calculator (enter last 4 weeks and get a forecast).	Streamlit, Plotly
Application layer	app.py receives the user's input, prepares the features and calls the models.	Python
Analysis layer	XGBoost regressor predicts next-week crime count; Random Forest classifier gives hotspot probability (High, Medium, Low risk).	scikit-learn, XGBoost
Data files	panel_train.csv (training data), next_week_input.csv (latest 4 weeks of each area) and meta.json (hotspot threshold and feature list).	CSV, JSON

9. Evaluation Plan

Item	Plan
Data split	Time-based: earlier weeks for training, later weeks for testing. No random shuffling, so future information never leaks into training. Three splits compared: 80:20, 70:30, 85:15.
Baseline	Predict next week's crime count as equal to last week's count (and as the 4-week average).
Regression metrics	MAE, RMSE and R-squared for the predicted weekly crime count per area.
Classification metrics	F1-score and ROC-AUC for hotspot / not-hotspot, with class weights because hotspots are the minority class.

Status: Done. All planned evaluations were carried out.

10. Outcomes

- A cleaned, documented dataset and a reproducible analysis notebook.
- Charts and maps that show when and where crime concentrates.
- A trained model that predicts crime level and hotspot status per area, with an honest comparison against baselines.
- A deployed web application that shows next-week hotspots on a map and lets a user check any location.
- Project documentation. **Status: Done.**

11. Limitations and Challenges

- Reported crime only: unreported crime is not in the data, and reporting rates differ between areas.
- Timestamp quality: records whose exact time is unknown may be stamped 00:00 or 12:00, so hour-wise findings are less reliable.
- One year of data: year-over-year trends and 'same period last year' features need more years.
- Sparse grid data: small grid cells may have many weeks with zero crimes, which can make prediction harder.
- Not Indian data: the models are trained on Chicago, so results describe Chicago and should not be read as predictions for Indian cities.
- Ethical care: predictions describe places, not people, and should support decisions, not replace human judgement.

12. References

- City of Chicago Data Portal. *Crimes - 2001 to Present*. data.cityofchicago.org (dataset id ijzp-q8t2).
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- Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of KDD 2016*.
- Ester, M. et al. (1996). A density-based algorithm for discovering clusters in large spatial databases with noise (DBSCAN). *Proceedings of KDD-96*.